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Cybersecurity by the Numbers [Infographic]

September 8, 2023 | Purdue Global

More than 1,800 data breaches impacted at least 422 million individuals in 2022, according to the Identity Theft Resource Center's **2022 Data Breach Report (https://www.idtheftcenter.org/wp-content/uploads/2023/01/ITRC_2022-Data-Breach-Report_Final.pdf)**. This is a slight decrease from the more than 1,860 data breaches that were reported in 2021.

According to the Stefanini Group (https://stefanini.com/en/insights/articles/cyber-security-statistics-for-2022-data-and-trends), cybercrime has increased 300% since the start of the pandemic. Data breaches in health care alone have increased by 58% during this time.

Demand for cybersecurity professionals continues to skyrocket. The **U.S. Bureau of Labor Statistics (https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm#tab-6)** (BLS) projects a 35% growth in the need for information security analysts from 2021 to 2031. In fact, the BLS projects 19,500 openings for information security analysts, on average, each year over the decade. The BLS credits this growth to a shift to remote work and the rise of e-commerce, including growth in digital health services and telehealth.

In this infographic, we explore the growing demand for cybersecurity professionals and the industry trends that drive such demand, including:

- Business costs associated with data breaches

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- The industries at the most risk for falling victim to cybercriminals
- The cybersecurity roles with the most employment potential
- The most in-demand certifications

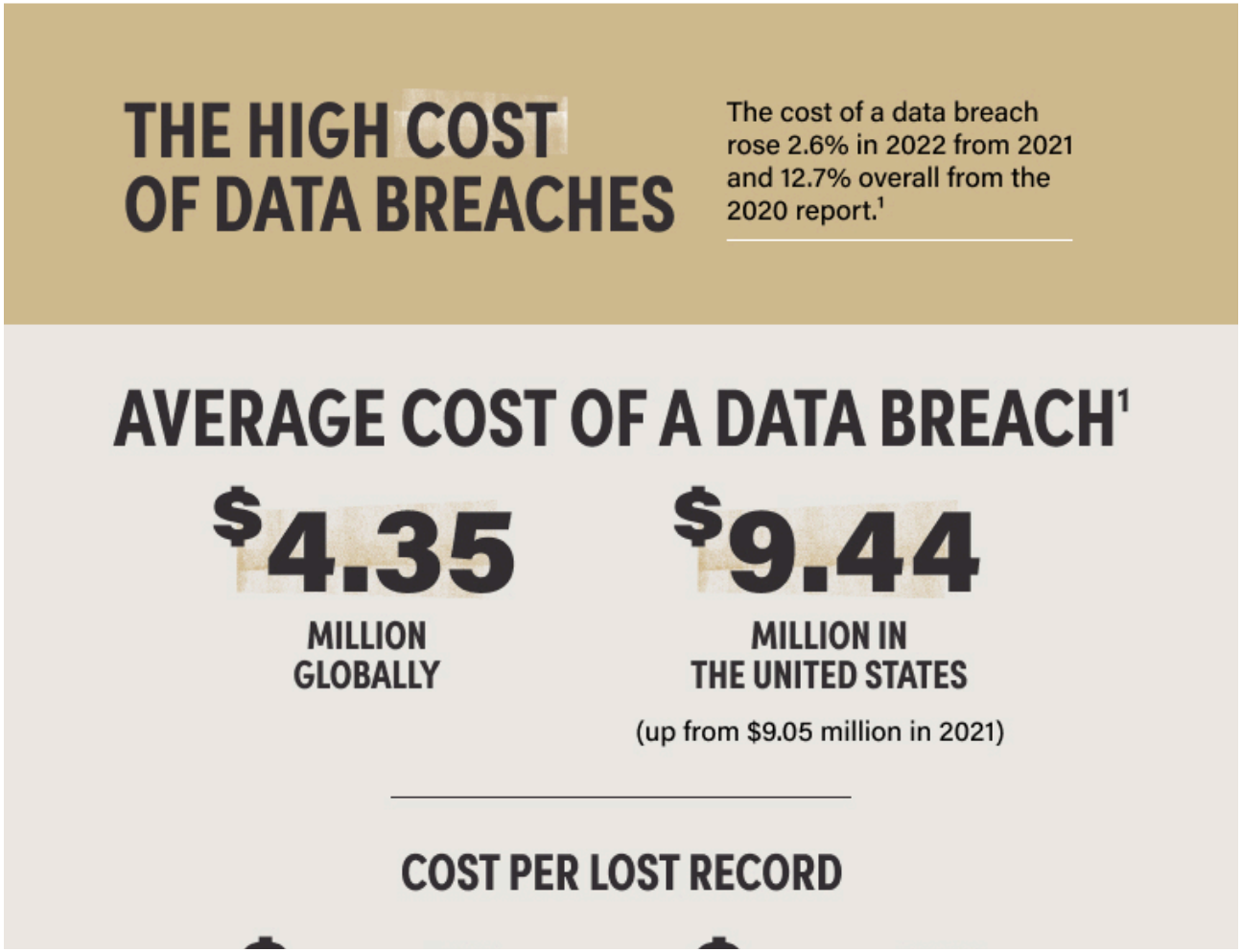
Cybersecurity remains one of the most promising industries in which to pursue a career, especially as demand shows no sign of slowing. With businesses relying more and more on data to drive revenue and growth, cybersecurity experts will

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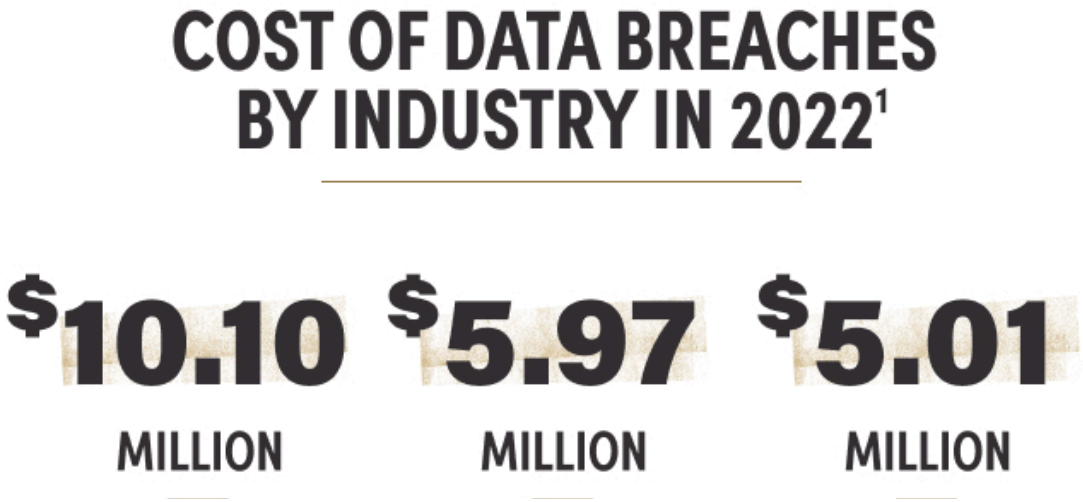
play an integral part in our economic future.

CYBERSECURITY By the Numbers

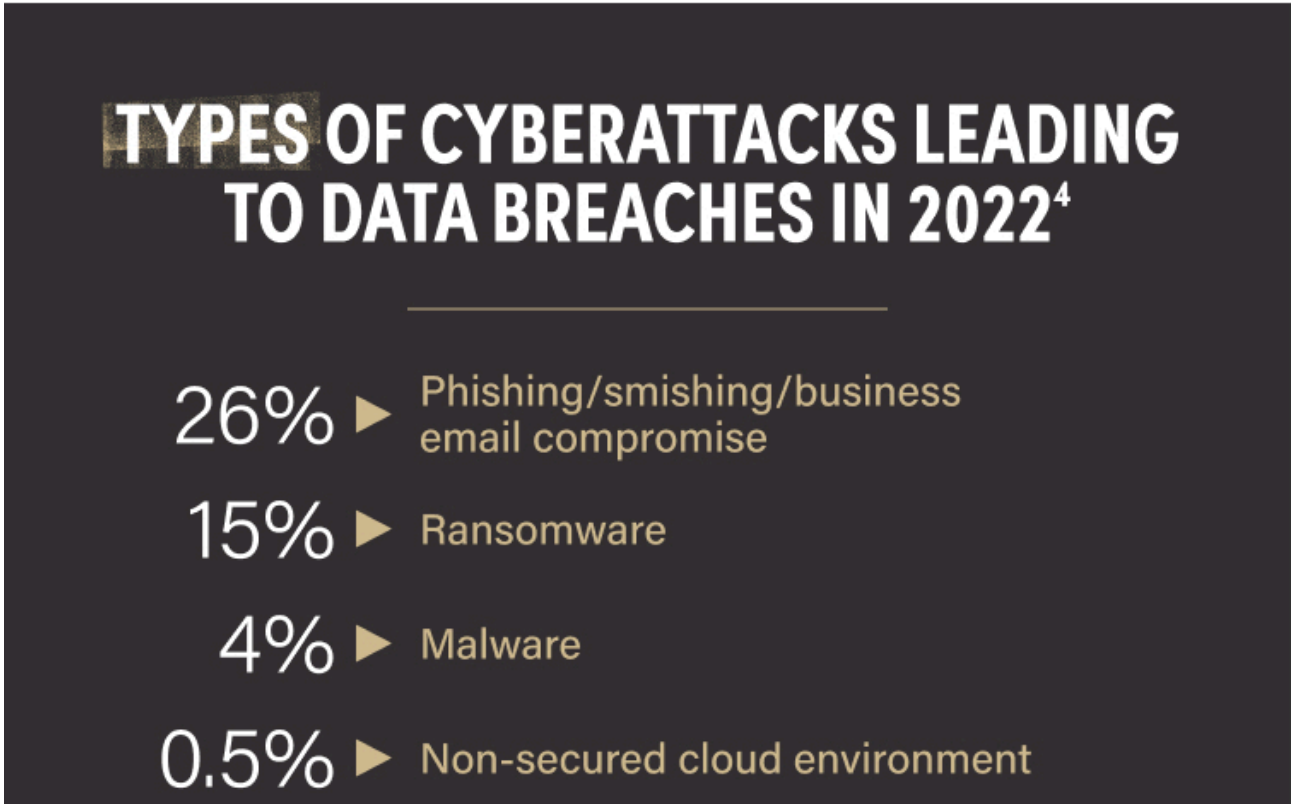


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$164
GLOBALLY
$242
IN THE UNITED STATES
12th Year in a Row:
THE U.S. WAS THE COSTLIEST COUNTRY FOR AVERAGE TOTAL DATA BREACH COST.[1]
ANNUAL COST
The annual cost of global cybercrime damages is estimated to increase to
$10.5 TRILLION BY 2025[3]


COST OF DATA BREACHES BY INDUSTRY IN 2022[1]
$10.10 MILLION
$5.97 MILLION
$5.01 MILLION

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TYPES OF CYBERATTACKS LEADING TO DATA BREACHES IN 2022[4]
26% Phishing/smishing/business email compromise
15% Ransomware
4% Malware
0.5% Non-secured cloud environment

1% ▶ Credential stuffing

0.4% ▶ Zero day attack

40% ▶ Not specified

1% ▶ Other

(Note: percentages are based on the total number of attacks for the year.)

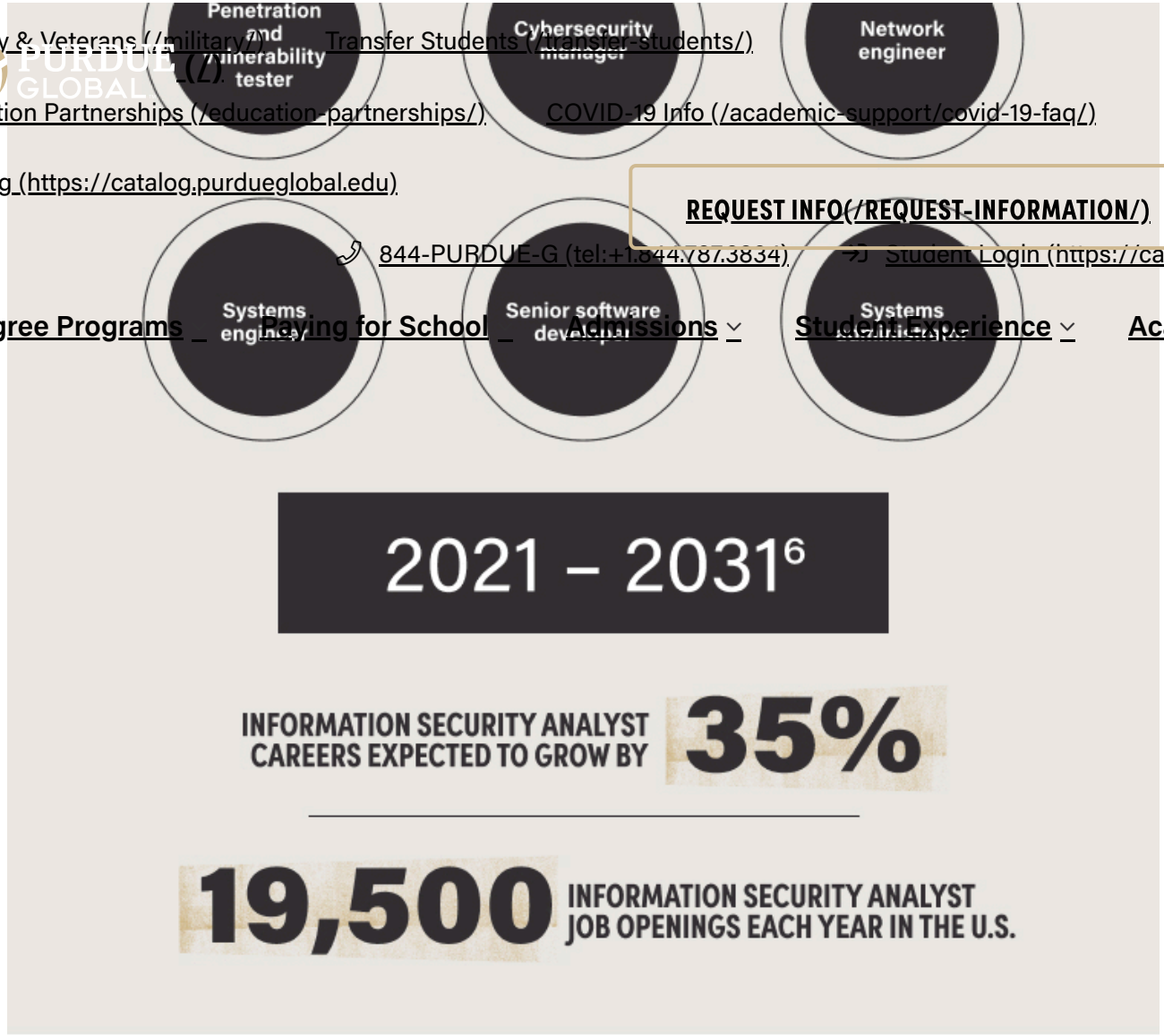
DEMAND FOR CYBERSECURITY PROFESSIONALS

Over 755,000 cybersecurity job openings were estimated to exist in 2022.[5]

TOP ROLES



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THE CYBERSECURITY SKILLS GAP[5]

Skills mentioned in job openings as categorized by the Workforce Framework for Cybersecurity:

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525,608
Operate and maintain

481,733
Securely provision

357,786
Oversee and govern

297,185
Protect and defend

205,780
Analyze

97,050
Collect and operate

58,672
Investigate

MOST IN-DEMAND CERTIFICATIONS[5]

Certified Information Systems Security Professional® (CISSP®) ▶ 134,613 Job Openings Requesting Certification

CompTIA Security+ ▶ 101,725 Job Openings Requesting Certification

Certified Information Systems Auditor® ▶ 87,519 Job Openings

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(/blog/it/cybersecurity-by-numbers-full.png)

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Cybersecurity by the Numbers [Infographic] Content

The High Cost of Data Breaches

The cost of a data breach rose 2.6% in 2022 from 2021 and 12.7% overall from the 2020 report.[1]

Average Cost of a Data Breach[1]

- $4.35 million globally
- $9.44 million in the United States (up from $9.05 million in 2021)

Cost per Lost Record

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- $164 globally[1]
- $242 in the United States[2]

12th year in a row, the US was the costliest country for average total data breach cost.[1]

Annual Cost

The annual cost of global cybercrime damages is estimated to increase to $10.5 trillion by 2025.[3]

Cost of Data Breaches by Industry in 2022[1]

- $10.10 million – Health care
- $5.97 million – Financial
- $5.01 million – Pharmaceutical
- $4.97 million – Technology
- $4.72 million – Energy

Types of Cyberattacks Leading to Data Breaches in 2022[4]

- 26% – Phishing/smishing/business email compromise
- 15% – Ransomware

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- 4% – Malware
- 5% – Non-secured cloud environment
- 1% – Credential stuffing
- 0.4% – Zero day attack
- 40% – Not specified
- 1% – Other

Note: Percentages are based on the total number of attacks for the year.

Demand for Cybersecurity Professionals

Over 755,000 cybersecurity job openings were estimated to exist in 2022.[5]

Top Roles

- Cybersecurity analyst
- Software developer
- Cybersecurity consultant
- Penetration and vulnerability tester
- Cybersecurity manager
- Network engineer
- Systems engineer

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- Senior software developer (7)
- Systems administrator (6)

From 2021 to 2031:

- Information security analyst careers expected to grow by 35%
- 19,900 information security analyst job openings each year in the U.S.

The Cybersecurity Skills Gap[5]

Skills mentioned in job openings as categorized by the Workforce Framework for Cybersecurity:

- Operate and maintain – 525,608
- Securely provision – 481,733
- Oversee and govern – 357,786
- Protect and defend – 297,185
- Analyze – 205,789
- Collect and operate – 97,050
- Investigate – 58,672

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Most In-Demand Certifications[5]

Certified Information Systems Security Professional (CISSP) – 134,615 job openings requesting certification

CompTIA Security+ – 101,725 job openings requesting certification

Certified Information Systems Auditor (CISA) – 87,519 job openings requesting certification

Certified Ethical Hacker (CEH) – 67,501 job openings requesting certification

Certified Information Security Manager (CISM) – 55,401 job openings requesting certification

Global Information Assurance Certification (GIAC) – 57,408 job openings requesting certification

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SOURCES:

1. **IBM/Ponemon Institute 2022 Cost of a Data Breach Report (https://www.ibm.com/reports/data-breach)**
2. **IBM 2019 Cost of a Data Breach Report (https://www.ibm.com/downloads/cas/RDEQK07R)**
3. **Cybersecurity Ventures (https://cybersecurityventures.com/cybercrime-damages-6-trillion-by-2021/)**
4. **Identity Theft Resource Center 2022 Data Breach Report (https://www.idtheftcenter.org/wp-content/uploads/2023/01/ITRC_2022-Data-Breach-Report_Final.pdf)**

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5. **CyberSeek Cybersecurity Supply and Demand Report (https://www.cyberseek.org/heatmap.html)**
6. **U.S. Bureau of Labor Statistics (BLS) (https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm#tab-1)**

ABOUT THE AUTHOR

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